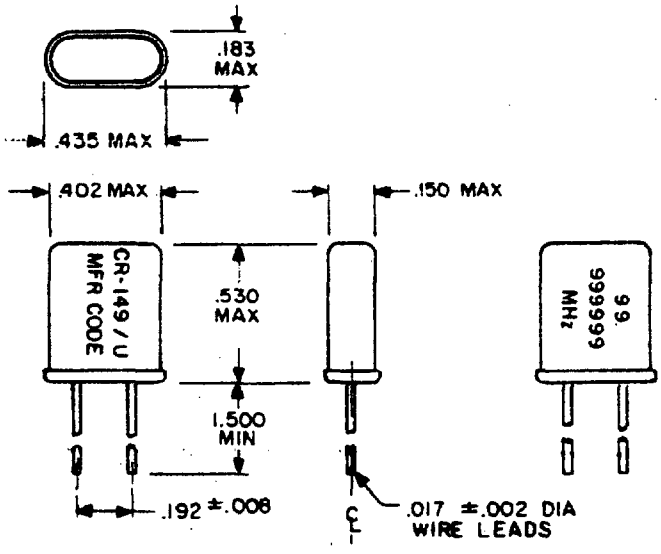
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-149/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of MIL-C-3098.

Pertinent characteristics: 46 to 100 MHz; fifth mechanical overtone; noncontrolled; series resonance.



INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.150	3.81	.402	10.21
.008	.20	.183	4.65	.435	11.05
.017	.43	.192	4.88	.530	13.46
				1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Dimensions and configuration.

Ⓐ denotes changes

FSC 5955

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:58Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U (MIL-H-10056/9)
Dimensions, marking and configuration	See figure
Frequency range	46 to 100 MHz, inclusive
Capacitance, shunt	6.0 pF ±0.5, -1.0 pF
Mode of oscillation	Fifth mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 mW, maximum
Calibration values:	
Resistance	40 ohms
Voltage drop	0.28 volt
Operating temperature ranges:	
Room temperature	25°C to 30°C, inclusive
Primary	10°C to 25°C and 30°C to 80°C, inclusive
Secondary	-40°C to +10°C and 80°C to 90°C, inclusive
Frequency tolerances:	
(A) Room temperature	±10 PPM
(A) Primary operating temperature range	±25 PPM
(A) Secondary operating temperature range	±50 PPM
Equivalent resistance	50 ohms, maximum
Shock:	
(A) Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
(A) Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
(A) Thermal shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
(A) Frequency change permitted	±5 PPM

1/ Method 204 of MIL-STD-202, test condition A.

NOTE: These crystal units are for use in the AN/ARC-101 equipment. They were previously fabricated using Bendix Corp. Specification 2088096-123 through 144.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army -
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5955-0512-25)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:58Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3098/130A CRYSTAL UNIT, QUARTZ, CR-149/U

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:58Z
Check the source to verify that this is the current version before use.